motion - practice problems

1. A small camera drone starts 2 m above the ground while filming a school field. It rises to 14 m above the ground, and the camera records for 9 s. What is its vertical displacement?

2. A student starts at a locker marked -12 m and walks to a classroom marked 35 m. What is the student's displacement?

3. A runner sprints 0.10 km east, turns around, and jogs 35 m west. What are the runner's distance traveled and displacement?

4. A city bus leaves the science museum stop at 7:48 AM and arrives at school at 8:13 AM. The route is listed as 6.2 km, but only the clock times matter here. How long was the ride in minutes?

5. A runner moves for 12 s and travels 80 m along a curved path. What is the runner's displacement?

6. A skateboarder goes from the 0.008 km mark to the 68 m mark in 15 s. What is the skateboarder's average velocity?

7. A dog runs 18 m east, then 6 m west, in 8 s. What are the dog's distance traveled and average velocity?

8. An elevator rises with an average velocity of 1.5 m/s. How long does it take to rise 36 m?

9. A train backs up slowly while coupling to another car. It moves at 43.2 km/h for 8 s. Let forward be positive, so the train's velocity is negative. What is its displacement?

10. A position vs. time graph goes through the points (2 s, 5 m) and (8 s, 29 m). What is the slope of the graph?

11. A position vs. time graph falls from 40 m at 0 s to -20 m at 10 s. What average velocity does the graph show?

12. A skateboarder speeds up from 1.5 m/s to 7.5 m/s in 4 s. What is the skateboarder's average acceleration?

13. A bike is moving forward at 9 m/s and comes to rest in 3 s. What is the bike's average acceleration?

14. A cart is moving left at 4 m/s, then 5 s later it is moving right at 6 m/s. Let right be positive. What is the cart's average acceleration?

15. A velocity vs. time graph goes through the points (3 s, 2 m/s) and (11 s, -6 m/s). What is the slope of the graph?

16. A cyclist on a straight bike path starts a timer at the 0.020 km mark and is at the 110 m mark after 0.50 min. The bike computer also reports that the cyclist's velocity changes from 2 m/s to 8 m/s. Find the displacement, average velocity, and average acceleration.

17. Two cars drive toward each other on a straight east-west road. Let east be positive. At the same instant, one car is at the 0 m marker and travels east at a constant 20 m/s. The other car is at the 600 m marker and travels west at a constant 10 m/s. How long until the cars meet?